Lending Game – Stage 1: the final pricing rule

Pricing the pool we actually win – Team "Lender 2"

Internal note; supersedes the band section of the stage-1 strategy note

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1 The rule, step by step

We are one of three lenders bidding on the same 100,000 applicants. Each applicant takes the cheapest offer (with a 2% preference for one lender), repays with interest or defaults and costs us the whole loan. Every lender sees the same public variables and *one private signal of its own*; the other two lenders' signals are information we do not have. Our rule prices each applicant in five steps.

Step 1 – estimate the probability of default. A gradient-boosted tree model, trained on the 100,000 past loans, gives $\hat{p}$ = the probability that the applicant defaults, from employment, marital status, income and our signal s_2 (a signal of exactly 0 is treated as "no signal"). It is well calibrated: among applicants with $\hat{p} = 10\%$, 10% default. Out-of-sample AUC 0.735.

Step 2 – the plain break-even rate. A loan at rate r earns $10{,}000\,r$ if repaid and loses 10,000 if not, so it breaks even when $(1-\hat{p})\,r = \hat{p}$, i.e. at $r = \hat{p}/(1-\hat{p})$: 5.3% for a 5% PD, 11% for 10%, 25% for 20%. This is the rate a lender would charge if the applicants who accept were a random sample.

Step 3 – the adjusted break-even: who actually accepts? They are not a random sample. Each rival sees a signal we do not; when that signal is bad the rival asks for more, and the applicant comes to us. So the applicants who accept our offer are riskier than $\hat{p}$, and the more so the riskier the segment. We measured this on the past loans, where all three signals are visible (Section 2): the default rate of the people *we win* is $\tilde{\pi}(\hat{p})$ – 12.7% for applicants we rate at 10%, 18% at 15%, 27% at 20%, 36% at 25%, above 50% beyond 27%. The *adjusted break-even* $\tilde{\pi}/(1-\tilde{\pi})$ is the rate at which winning these applicants still breaks even: 14.5% at $\hat{p} = 10\%$ instead of 11%, 37% at 20% instead of 25%, and no finite rate beyond 27%.

Step 4 – the margin. On safe applicants (up to a threshold τ of 8–12% PD at low income, 25% above 20k of income, Table 1) the rivals compete within a couple of points and the 2% preference means we only keep our own third of the market if we stay close to them: the margin is 3–4 points over the plain break-even, which is about 2 points over the adjusted one. Above the threshold the rivals progressively stop competing (their own thresholds lie between 8% and 20%), their prices disperse, and a larger margin captures more surplus without losing share: the margin rises smoothly to $X = 5$ points over the adjusted break-even, the rise being centred one point above the threshold and spread over about 8–14% PD. $X = 5$ points was chosen by simulation (Section 3): a larger X earns more when we are the cheapest but is undercut more often by rivals pricing the same way.

Step 5 – the cap. No rate exceeds 0.989. In this game an applicant who receives at least one offer always borrows, so a capped offer is a harmless "lender of last resort" quote: it is taken only if both rivals refuse the applicant, and 0.989 beats a rival who caps at the obvious 0.99 or 1.00. It concerns 2% of applicants (PD above ∼27%), for whom no finite competitive rate exists.

The formula. With s the income segment, $m(s)$ and $\tau(s)$ from Table 1,

$$r_{\text{safe}}(\hat{p}) = \frac{\hat{p}}{1-\hat{p}} + m(s), \qquad r_{\text{band}}(\hat{p}) = \min\Big(\frac{\tilde{\pi}(\hat{p})}{1-\tilde{\pi}(\hat{p})} + X,\ 0.989\Big), \qquad X = 0.05, \tag{1}$$

$$r(\hat{p}) = r_{\text{safe}} + w(\hat{p})\,\big(r_{\text{band}} - r_{\text{safe}}\big), \qquad w(\hat{p}) = \frac{1}{1+\exp\big(-(\hat{p}-\tau(s)-0.01)/0.01\big)}. \tag{2}$$

w is a logistic weight: about 0.05 two points below the threshold, 0.5 one point above it, 0.95 three points above. Rates are rounded to four decimals; every applicant receives an offer.

Income segment	0	1–5k	5–10k	10–20k	20–40k	40k+
margin $m(s)$ on the safe part	4.0%	3.5%	3.5%	3.0%	3.0%	3.0%
threshold $\tau(s)$	12%	12%	10%	8%	25%	25%
share of the segment above the threshold (new file)	92%	74%	69%	18%	0%	0%

Table 1: Segment parameters, chosen by maximizing expected profit over competitor scenarios on the past loans. Thresholds are low where the rivals' hidden information is large (standard deviation of $p_{\text{full}} - \hat{p}$: 7 points at zero income, 0.3 points above 40k).

PD $\hat{p}$	plain break-even	adjusted break-even	margin over adjusted	**rate offered**	what happens
6%	6.4%	7.5%	2.4 pts	**9.9%**	competitive pricing, we keep our fans
8%	8.7%	10.6%	1.8	**12.4%**	idem
10%	11.1%	13.7%	2.0	**15.7%**	threshold: the margin starts to rise
12%	13.6%	16.4%	3.9	**20.3%**	rivals begin to drop out
14%	16.3%	21.2%	4.7	**25.9%**	
16%	19.0%	27.1%	4.9	**32.0%**	full margin: adjusted break-even + 5
20%	25.0%	36.1%	5.0	**41.1%**	
25%	33.3%	64.7%	5.0	**69.7%**	the selected pool defaults 36%
30%	42.9%	none	–	**98.9%**	cap: taken only if both rivals refuse

Table 2: The rule at a glance, for an applicant with income 5–10k (threshold 10%, margin 3.5%). Example: income 7,000 €, PD 12%: plain break-even 0.12/0.88 = 13.6%; the applicants we would win at that PD default 14.1%, so the adjusted break-even is 0.141/0.859 = 16.4%; the margin at 12% PD is 3.9 points; offer 20.3%.

2 Where $\tilde{\pi}$ comes from: the break-even given that we win

The measurement. For every past loan i we compute the three lenders' out-of-sample estimates $\hat{p}_{k,i} = \mathbb{E}[D \mid X, s_k]$, $k = 1,2,3$ (each with its own signal), and the full-information probability $p_i^{\text{full}} = \mathbb{E}[D \mid X, s_1, s_2, s_3]$. We let every lender quote $\hat{p}_{k,i}/(1-\hat{p}_{k,i}) + m(s_i)$ – the same rule, each on its own information – and apply the choice rule of the game for each applicant type $t \in \{1,2,3\}$ (one third each; the preferred lender's rate counts 0.02 lower):

$$\omega_i \;=\; \frac{1}{3}\sum_{t=1}^{3} \mathbf{1}\Big\{\, r_{2,i} - 0.02\cdot\mathbf{1}[t=2] \;<\; \min_{k\neq 2}\big(r_{k,i} - 0.02\cdot\mathbf{1}[t=k]\big)\Big\} \tag{3}$$

is the probability that applicant i takes our offer ("share we win" in Table 3 is its average). For each PD bin B (one point wide),

$$\tilde{\pi}(B) = \frac{\sum_{i\in B}\omega_i\, p_i^{\text{full}}}{\sum_{i\in B}\omega_i}, \qquad \tilde{\pi}^{\text{real}}(B) = \frac{\sum_{i\in B}\omega_i D_i}{\sum_{i\in B}\omega_i}, \qquad \tilde{\pi}^{95}(B) = \tilde{\pi}^{\text{real}} + 1.645\sqrt{\frac{\tilde{\pi}^{\text{real}}(1-\tilde{\pi}^{\text{real}})}{\sum_{i\in B}\omega_i}}. \tag{4}$$

$\tilde{\pi}$ is the default rate of the applicants we win; the expected version (with p^{full}, smooth) is used in the rule, the realized version and its 95% upper bound check it (Table 3, Figure 1A).

Our PD $\hat{p}$	Share we win	$\tilde{\pi}$ (expected)	$\tilde{\pi}^{95}$ (realized, upper)	Penalty $\tilde{\pi} - \hat{p}$	Plain break-even	Adjusted break-even
5%	44%	5.1%	5.5%	+0.6 pt	4.7%	5.3%
8.5%	39%	10.2%	11.3%	+1.7	9.2%	11.3%
10.5%	38%	12.7%	14.2%	+2.1	11.7%	14.5%
12.5%	44%	14.4%	15.8%	+1.9	14.3%	16.9%
14.5%	29%	18.2%	19.9%	+3.7	16.9%	22.2%
16.5%	17%	22.9%	26.7%	+6.4	19.7%	29.6%
18.5%	18%	24.1%	27.9%	+5.6	22.7%	31.7%
20.5%	11%	26.9%	30.1%	+6.5	25.7%	36.8%
22.5%	18%	29.0%	37.0%	+6.5	29.1%	40.8%
24.5%	13%	35.5%	39.7%	+11.0	32.5%	55.0%
26.5%	4%	46.8%	58.2%	+20.2	36.2%	88.1%
$\geq 28\%$	$\leq 1\%$	$> 50\%$	–	> 20	$\geq 40\%$	none below the cap

Table 3: The winner's curse per PD level, measured on the past loans when every lender prices on its own model.

Three remarks. (i) The selection, hence $\tilde{\pi}$, is the same under *any* common increasing transformation of the rule: if every lender applies the same increasing function to its own PD, the same applicants are won. The curve therefore does not depend on the margin the rivals use – only on their models being about as informative as ours. (ii) The curse is small where our model can compete (below 10% PD the penalty is at most 2 points) and explodes above 25%: beyond ∼27% the applicants we would win in competition default more than half the time and no finite rate breaks even. (iii) The 95% upper bound stays within 1–4 points of the expected curve up to 25% PD, so the five-point margin also covers the estimation uncertainty there.

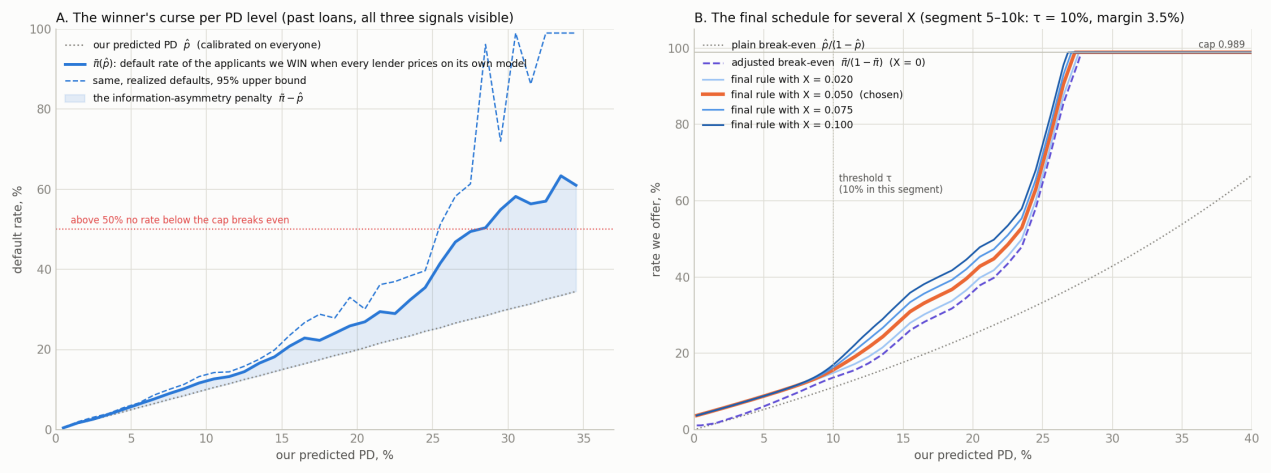


Figure 1: A: our predicted PD versus the default rate of the applicants we win (expected, and realized 95% upper bound); the shaded area is the price of the information asymmetry. B: the final schedule in a typical low-income segment (threshold 10%, margin 3.5%) for several margins X; $X = 0.05$ is the submitted rule.

3 Choosing the margin X and its transition

3.1 The level of X: a Monte Carlo over rival strategies

At $X = 0$ we would break even on the band even if we won all of it; the margin is what we earn there. But every point of margin is a point by which a rival pricing the same way can undercut us, and with the 2% preference whoever is lowest within two points takes the whole band. So X is a bet on the rivals' margins. Each rival is drawn at random from a profile of plausible behaviours and priced on its own PD estimate; the market is evaluated exactly (three applicant types with weight 1/3, ties split evenly – an adverse-tie variant gives identical results, since the schedules create no

exact ties), rates rounded as in the file, profits computed with p^{full}. Profiles and weights (our belief that the rivals are curse-aware and do not use a lender-of-last-resort rate): *like us, no net* – rivals pricing the band at their own adjusted break-even plus a margin drawn between 3 and 25 points (40%); *smart* – curse-aware rivals using a net, a ramp, a large margin, a data-driven schedule, refusal or a compromise (20%); *clones* of our method with other parameters (10%); *best-responders* who undercut our schedule (10%); a broad *mixed* profile including naive behaviours (10%); pure *netters* (5%); *model-followers* pricing model + 2–6% for everyone (5%). 120 independent rival pairs per profile.

Band schedule	Prior-weighted over the seven profiles mean profit	P(first)	P(last)	"Like us, no net" rivals only mean profit	P(first)	P(last)
adjusted break-even, $X = 0$	11.9 M€	79%	2%	9.7	62%	4%
$X = 0.02$	13.6	82%	1%	11.0	70%	2%
$X = 0.05$ **(submitted)**	**15.7**	**75%**	**3%**	**12.4**	**55%**	**6%**
$X = 0.075$	17.1	70%	5%	12.9	47%	11%
$X = 0.10$	18.3	65%	7%	13.1	36%	17%
$X = 0.15$	19.6	54%	15%	12.2	14%	35%
ramp 0.45 → 0.65 (previous candidate)	22.2	42%	41%	8.2	2%	91%
flat net 0.828 (first file)	11.0	13%	68%	6.7	0%	98%

Table 4: Expected profit and rank among the three lenders as a function of the margin X (120 random rival pairs per profile). Profit rises with X; the probability of finishing first falls with it.

The criterion. Stage-1 performance is graded on the level of profit *and* on the ranking, so we declared the rule: maximize the prior-weighted expected profit subject to a probability of finishing last of at most 3%. It selects $X = 0.05$: 15.7 M€ on average, first three times out of four, last 3% of the time, never below 7.2 M€ on average in any profile. Higher margins buy profit at the cost of rank (at $X = 0.10$, 18.3 M€ but last 7% – 17% against rivals who price like us); the hand-made ramp earns more on average only because of jackpots against netters and finishes last in 41% of the draws (91% against rivals who price like us). Figure 2 shows the dial.

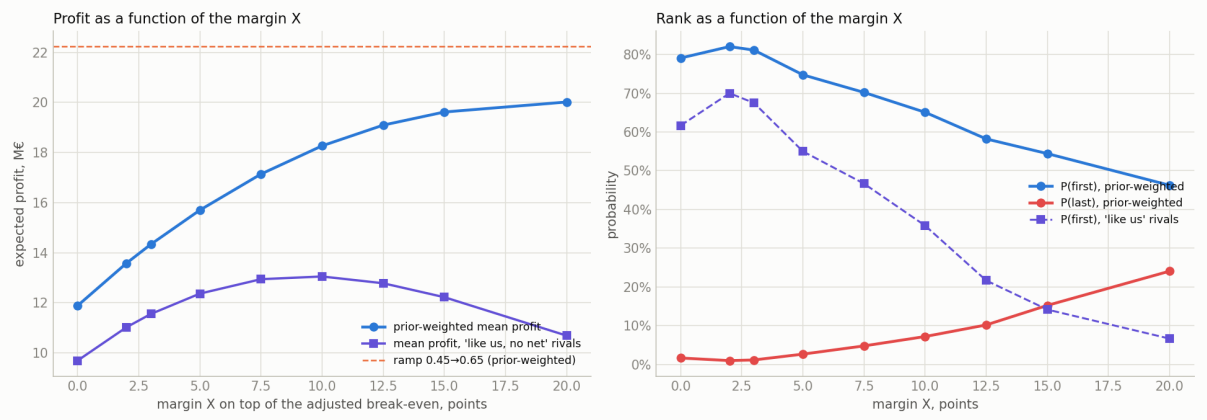


Figure 2: Expected profit (left) and rank (right) as a function of the margin X on top of the adjusted break-even.

3.2 The transition: where the margin rises

A first version switched from the safe-part margin to the band margin in one point of PD at the threshold, which produced a jump of five points in the rate between 10% and 11% PD (14.8% to 19.7% in the 5–10k segment) although the default rate of the people we win moves by one point there – an artefact of gluing two formulas, with no economic event behind it. We therefore let the margin rise smoothly, and chose the width by the same Monte Carlo:

Transition of the margin	rate at 10 / 11 / 12 / 14% PD	prior-weighted mean	P(first)	P(last)
one-point step at the threshold	14.8 / 19.7 / 21.4 / 26.2	16.3 M€	82%	1%
logistic, centre τ+1 pt, scale 1 pt (submitted)	**15.7 / 17.9 / 20.3 / 25.9**	**15.8**	**82%**	**1%**
logistic, centre τ+2, scale 1.5	15.4 / 17.2 / 19.3 / 24.9	15.2	83%	2%
logistic, centre τ+3, scale 2.5	15.6 / 17.1 / 18.9 / 23.6	14.4	83%	2%

Table 5: Smoothing the margin transition (100 random rival pairs per profile, segment 5–10k for the rates).

The submitted transition (spread over roughly 8–14% PD) costs 0.5 M€ of expected profit – inside the simulation's noise – and nothing in rank; wider transitions give margin away to applicants most rivals have already stopped competing for. Economically the width matches where rivals disengage: their thresholds, in every profile we consider, lie between 8% and 20% PD and concentrate around 10–12%.

3.3 The cap: are the capped offers profitable?

Above ∼27% PD the adjusted break-even exceeds 100%, so the rule caps the rate at 0.989 for about 2% of applicants (2,027 in the submitted file; 1,946 of the past loans on which the simulations below run). Does that mean we lose money on them? No – the two statements concern two different pools. The adjusted break-even is the break-even of the applicants we would win *in competition*, i.e. because our estimate is the lowest of the three; above 27% PD that pool defaults more than half the time, which is exactly why the rule stops competing there. At 0.989 we are never the lowest finite quote: a capped offer is taken only when *both rivals refuse or cap* the applicant, a pool selected by the rivals' refusal rules, not by our estimate being the lowest. Table 6 measures it: where we win any capped loans their expected default is 33–46%, comfortably below the 50% at which a 98.9% loan breaks even; the one profile with a worse pool (model-followers, 59%) yields two loans. Prior-weighted, keeping the cap earns +0.10 M€, refusing those applicants 0, capping only up to 30% PD +0.06 M€. The cap is kept: a small, never materially negative, lender-of-last-resort option.

Rival profile	expected loans won at the cap (of 1,946 offers)	their expected default	profit on them
like us, no net	122	46%	+0.10 M€
smart	70	33%	+0.23
mixed	71	40%	+0.14
netters	26	33%	+0.09
clones	3	39%	+0.01
best-responders	0	–	0
model-followers	2	59%	−0.003

Table 6: The capped offers by rival profile (exact types, fair ties, 60 rival pairs per profile).

4 The submitted file

`output/round1/rates_round1.csv`, built from the model refitted on all past loans (full-precision PDs) by

```
python make_submission.py --adjusted-x 0.05 --cap 0.989 --phase-delta 0.01
                          --phase-width 0.01
```

100,000 rows, columns `id` and `rate`, ids 1 to 100,000 in order and unique, no missing value, all rates in $[0.0304, 0.989]$; the SHA-256 checksum (`5aba4d61...`) and the exact parameters are stored in `rates_round1_params.json`.

rate	≤5%	5–10%	10–15%	15–20%	20–30%	30–40%	40–50%	50–60%	60–80%	80–98.9%
offers	13,474	25,593	13,596	12,482	12,854	10,219	3,576	3,559	1,515	3,132

Table 7: Composition of the file: 65,145 applicants below their segment's threshold, 45,582 priced on the adjusted break-even (12.8% to 98.9%), of which 2,027 at the cap.

5 What we expect, and what could go wrong

Across the rival profiles the expected profit of this rule is 7–16 M€ (median round 6–8 M€), finishing first in about three quarters of the simulated markets. The two assumptions that matter most are (i) that the rivals' models are about as informative as ours – if theirs are worse, the curse is smaller and we are slightly over-pricing the band; if better, the five-point margin is our cushion – and (ii) the rules as written: an applicant with at least one offer always borrows, which is what makes the capped offers harmless. After stage 1 we will read the rivals' actual band prices from the results and re-price each PD category at their observed level minus a few points where that stays above the adjusted break-even, which is the only part of the rule that depends on a belief rather than on a measurement.